

EDUCATION

- University of California, Berkeley** Berkeley, California
 • *PhD Candidate in Electrical Engineering & Computer Science. Member of BAIR lab.*
Advisors: Prof. Michael (Miki) Lustig and Prof. Stella Yu. GPA: 4.0 / 4.0. *Aug. 2020 – 2025*
- Pontificia Universidad Católica de Chile (PUC)** Santiago, Chile
 • *Bachelor's Degree in Engineering.*
Electrical Engineering Major, Computer Science Minor.
GPA: 6.6 / 7.0. Ranked No.1 out of 875 students in my graduating class. *Mar. 2014 – Jul. 2018*
- University of California, Davis** Davis, California
 • *UCEAP Exchange Program, Computer Science & Engineering Major.*
GPA: 4.0 / 4.0. Fall and Winter quarters. Dean's Honor List. *Sep. 2017 – Mar. 2018*

WORK EXPERIENCE

- Adobe Inc.** San Jose, California
 • *Research Scientist/Engineer Intern*
Research on large language models as dynamical systems. *May. 2023 – Dec. 2023*
- Google LLC** Sunnyvale, California
 • *PhD Software Engineer Intern*
Research on deep learning for differentially private generative models. *May. 2022 – Aug. 2022*
- HeartVista Inc.** Los Altos, California
 • *Machine Learning Software Engineer*
Develop clinical level deep learning models for the first FDA-cleared self-driving MRI. *Jul. 2018 – Aug. 2020*

TEACHING & RESEARCH

- Graduate Student Researcher** Berkeley Artificial Intelligence Research (BAIR)
 • *Advisors: Prof. Michael (Miki) Lustig and Prof. Stella Yu.*
Learning, Robotics, and Control for Rapid and Robust MRI. EECS Department, UC Berkeley
2020 - Present
- Graduate Student Instructor** EECS Department, UC Berkeley
 • *CS184/284A: Computer Graphics*
EECS16A: Designing Information Devices and Systems I *Spring 2024*
Fall 2022
- Research Assistant – Deep Learning Explainability** CS Departments, PUC & Stanford
 • *Advisors: Prof. Álvaro Soto, Ph.D. and Prof. Juan Carlos Niebles, Ph.D.* *2018 - 2019*
- Computer Science Tutor** UC Davis CS Tutoring, UCD
 • *Tutor for CS courses. Weekly tutoring office hours and occasional review sessions.* *Winter 2018*
- Research Assistant – Sonification of 4D Flow MRI** Biomedical Imaging Center, PUC
 • *Research Advisor: Prof. Pablo Irarrázaval, Ph.D.* *2016 - II*
- Project Assistant** Physics Department, PUC
 • *Supervisor: Prof. Juan Pedro Ochoa-Ricoux, Ph.D.*
Project: Facilitating the Understanding of University Physics through Class Experiments. *2016 - I*
- Teacher Assistant – Various Courses** PUC
 • *Prepare and lead weekly discussions sessions. Took, on average, 2 positions per semester.*
Courses in Electrical Engineering, Computer Sciences, Physics, and Mathematics Departments. *2015 – 2018*

PUBLICATIONS & PRESENTATIONS

- **Simultaneous Off-Resonance Correction and Fat-Water Separation From Center-Out Spiral Acquisition Using a Physics-Informed DL Framework:** A. De Goyeneche, S. Ramachandran, K. Wang, E. Karasan, J. Cheng, S. Yu, M. Lustig. In proceedings of the 33rd ISMRM Annual Meeting and Exhibition, Singapore, 2024 (Oral).
- **ResoNet: Noise-Trained Physics-Informed MRI Off-Resonance Correction:** A. De Goyeneche, S. Ramachandran, K. Wang, E. Karasan, J. Y. Cheng, S. X. Yu, M. Lustig. In *Advances in Neural Information Processing Systems (NeurIPS)*, New Orleans, 2023.
- **K-band: Self-supervised MRI Reconstruction via Stochastic Gradient Descent over K-space Subsets:** F. Wang, H. Qi, A. De Goyeneche, R. Heckel, M. Lustig, E. Shimron. Arxiv pre-print.
- **Physics Informed Deep Learning based Off-Resonance Correction Trained on Synthetic Data:** A. De Goyeneche, S. Ramachandran, K. Wang, E. Karasan, S. Yu, M. Lustig. In proceedings of the 31st ISMRM Annual Meeting and Exhibition, UK, 2022 (Oral)
- **Rigorous Uncertainty Estimation for MRI Reconstruction:** K. Wang, A. Angelopoulos, A. De Goyeneche, A. Kohli, E. Shimron, S. X. Yu, J. Malik, M. Lustig. In proceedings of the 31st ISMRM Annual Meeting and Exhibition, UK, 2022 (Oral)
- **BladeNet: Rapid PROPELLER Acquisition and Reconstruction for High spatio-temporal Resolution Abdominal MRI:** E. Shimron, A. De Goyeneche, K. Wang, A. B. Syed, S. Vasanawala, M. Lustig. In proceedings of the 31st ISMRM Annual Meeting and Exhibition, UK, 2022 (Oral)
- **High Fidelity Deep Learning-based MRI Reconstruction with Instance-wise Discriminative Feature Matching Loss:** K. Wang, J. I. Tamir, A. De Goyeneche, U. Wollner, R. Brada, S. Yu, M. Lustig. *Magnetic Resonance in Medicine*, 2022.
- **Deep-Learning-Based Motion Correction For Quantitative Cardiac MRI:** A. De Goyeneche, S. Tang, N. O. Addy, B. Hu, W. Overall, J. M. Santos. In proceedings of the 30th ISMRM Annual Meeting and Exhibition, Online, 2021 (Oral)
- **Explaining VQA predictions using Visual Grounding and a Knowledge Base:** F. Riquelme*, A. De Goyeneche*, Y. Zhang, J. C. Niebles, A. Soto. Journal publication at *Image and Vision Computing*. 2020. (*Equal contribution.)
- **One-Click Spine MRI:** A. De Goyeneche, E. Peterson, J. J. He, N. O. Addy, J. M. Santos. Abstract and poster presentation at *Medical Imaging meets NeurIPS Workshop*. Vancouver, Canada. 2019.
- **Automated Cardiac Magnetic Resonance Imaging:** A. De Goyeneche, N. O. Addy, J. M. Santos, B. Hu. Abstract and poster presentation at *American Heart Association Scientific Sessions*. Philadelphia, PA. 2019.
- **Rapid Automated Cardiac Imaging:** A. De Goyeneche, N. O. Addy, E. Peterson, H. Islam, W.R. Overall, J. M. Santos, B. Hu. Abstract and poster presentation at *Frontier of AI-Assisted Care (FAC) Scientific Symposium*. Stanford, CA. 2019.

AWARDS & FELLOWSHIPS

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| • EECS Undergraduate Researcher Mentoring Award
<i>Award for serving as undergraduates research mentor.</i> | UC Berkeley
2023 & 2024 |
| • EECS Departmental Fellowship
<i>Full tuition and fees plus stipend.</i> | UC Berkeley
Fall 2020 |
| • Academic Excellence Graduation Award
<i>Distinguished as rank 1st scholar of the Engineering B.S. graduation promotion at PUC.</i> | PUC
Oct. 2018 |
| • Dean's Honors List, College of Engineering
<i>Distinguished in both quarters of my exchange program.</i> | UC Davis
Fall 2017, Winter 2018 |
| • Highest Score in Fundamentals of Engineering Examination
<i>Tested in Feb. 2017 at PUC (1st out of 530 students).</i> | PUC
Apr. 2017 |
| • Engineering Honours Scholarship – Matrícula de Honor
<i>Distinguished as top scholar of the engineering promotion 2014 in PUC. (1st out of 875 students)</i> | PUC
Mar. 2017 |
| • Engineering Honours Scholarship – Matrícula de Honor
<i>First year 100% Scholarship because of Entry Tests Scores – PSU.</i> | PUC
Mar. 2014 |
| • National Admissions Test Double Award
<i>Doble Puntaje Nacional. Perfect Score in Math and Science Tests for Chilean college entry National Examinations (top 0.002% of total applicants). Equivalent to SAT.</i> | Government of Chile
Dec. 2013 |